

Rajashik Datta

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🔗 [Google Scholar](#)

🌐 [LinkedIn](#)

🐙 [Github](#)

🌐 [Portfolio](#)

Research Interests

Trustworthy ML (xAI + robustness), Computer Vision (Hyperspectral/Multimodal), Human-Computer Interaction

Education

Institute of Engineering & Management, Kolkata, India

August 2022 – July 2026

B.Tech in Computer Science & Engineering (Artificial Intelligence)

CGPA: 9.19 / 10

Ranked 6th among the top 10% of class by CGPA in Year 3 (AY 2024–25).

Awarded the **Chancellor's Award for Exemplary Research Contribution 2026**, the sole recipient in the department.

Experience

University of Nebraska-Lincoln, USA

June 2025 – Present

Research Intern

Remote (USA)

Supervisor: *Dr. Sruti Das Choudhury (Offer Letter)*

- Spearheaded an explainable AI + data-storytelling clustering pipeline across precision agriculture and pediatric health-care—grouping 22 Indian crop types using 7 agro-climatic/soil features and segmenting a 500-record hospital cohort—showing that z-score rescaling + removing binary gender prevents charge-dominated clusters and surfaces clinically meaningful cohorts (LOS up to 29 days; charges up to 34,644) for decision support.
- Developed a temporal-embedding visual analytics system for 42 plants from 9 genotypes over 25 days, engineering multi-scale phenotype descriptors (growth rates/accelerations, fourier spectra, wavelet energies, distributional stats) and achieving genotype-aligned DTW clustering (ARI 0.30; NMI 0.62) with cross-validated early-prediction curves and SHAP/LIME-linked causal graphs to explain when/why genotypes diverge.
- Implemented an interactive hyperspectral analysis tool, HyperProbe for calibrated datacubes spanning 517-1700 nm (B=243 bands), enabling rapid pixel/ROI annotation, band-difference + Otsu segmentation (IoU/F1 evaluation), and full-scene classification via 3 model families (MLP/logistic regression/random forest) with built-in ablations that log clicks/ROIs under fixed 5-min label budgets to quantify accuracy-per-effort.
- Featured in the university's news story for research contributions: snr.unl.edu (August, 2025)

University of Calcutta

January 2025 – Present

Research Scholar

Kolkata, India

Supervisors: *Dr. Arup Kumar Chattopadhyay, Prof. Amit Kumar Das, Prof. Amlan Chakrabarti*

- Engineered FHFAM (FH-FAM), a fuzzy-hypergraph feature selection algorithm, achieving the best mean accuracy (81.43%) and best mean feature reduction (89.28%) across 15 agriculture/remote-sensing datasets (5/15 wins) with 11.08s average runtime and statistically significant accuracy gains over key baselines (Wilcoxon $p < 0.05$).
- Proposed SIFHFAM, a stage-wise intuitionistic-fuzzy hypergraph selector with a monotone submodular coverage objective and greedy (1–1/e) guarantee, delivering the top average accuracy ($\approx 84\%$) while pruning $\approx 99\%$ features (typically retaining $< 2\%$) across 14 high-dimensional benchmarks in $\sim 0.1s$ /run under $10\times$ repeated 75/25 train-test splits.

Generative AI Centre of Excellence, IEM

November 2024 – Present

Student Research Lead; Senior Research Advisor at GenAI CoE

Kolkata, India

Established and led GenAI CoE's research sub-committee, end-to-end research execution and operations—recruited and onboarded members via interviews, mentored and staffed project teams, coordinated 10+ journal groups, maintained the CoE website, and launched *ReelBook* (Pearson collaboration) and *Medium publishing* to scale institute-wide research output and AI upskilling at IEM.

IEM Research Foundation

August 2024 – March 2025

Project Intern at bair.ai (Certificate)

Kolkata, India

Built *MemeMetric*, an end-to-end cluster-based cryptocurrency forecasting system by architecting the full data/ML pipeline with automated reporting, and integrated real-time Twitter/Telegram/Reddit sentiment signals via NLP to improve robustness and reduce forecast error/volatility.

Innovation & Entrepreneurship Development Cell (CSE), IEM

March 2024 – August 2024

Research Assistant (Certificate)

Kolkata, India

Co-authored an IEM-HEALS 2024 accepted study analyzing Jul 2019–Dec 2022 price dynamics of 20 pharma stocks using multivariate regression, volatility modeling, and event-study methods, and engineered *TraderBot*, a Flask+MongoDB real-time trading simulator wired to Yahoo Finance for live strategy backtesting and portfolio experiments.

National University of Singapore (NUS), Singapore

July 2023 (1 week)

Study Abroad Program (Certificate)

Singapore

Studied fundamentals of “Artificial Intelligence, Internet of Things, Machine Learning & Data Analytics”, lectured by *Dr. Peter Leong, Dr. Eric Cambria, Dr. Matthew Chua, Dr. Yiliang Zhao, Dr. Gábor Benedek, Dr. Tan Kian Hua, Yong Heng Michael Tan, Marton Szel, Gillian Cheng*.

Publications

Published/Accepted

1. **Rajashik Datta**, Sanjan Baitalik, “*Confidence as a Tie-Breaker: Reassessing Multilingual Hedging Bias in LLM-as-a-Judge Evaluation*”, The Association for Computational Linguistics (ACL Student Research Workshop), 2026.
2. Sanjan Baitalik, **Rajashik Datta**, “*Garden Path Recovery in Causal and Masked Language Models*”, The Association for Computational Linguistics (ACL Student Research Workshop), 2026.
3. Sanjan Baitalik, **Rajashik Datta**, Utsho Banerjee, Rajarshi Karmakar, Vincent Stoerger, Himadri Nath Saha, Sruti Das Choudhury, “*ReproPheno and ReproPhenoNet: A Large-Scale Multimodal Benchmark Dataset and Deep Learning Framework for Reproductive-Stage Plant Phenotyping*”, The Association for the Advancement of Artificial Intelligence (AAAI Workshop on AgriAI), 2026.
4. **Rajashik Datta**, Sanjan Baitalik, Amit Kumar Das, Sruti Das Choudhury, “*PlantPhenoLM: Phenotype-Genotype Mapping Inference with Multi-Turn LLM Reasoning and Selective Prediction*”, The Association for the Advancement of Artificial Intelligence (AAAI Bridge on Logic & AI), 2026.
5. Sanjan Baitalik, **Rajashik Datta**, Amit Kumar Das, Sruti Das Choudhury, “*Conversation as Belief Revision: GreedySAT Revision for Global Logical Consistency in Multi-Turn LLM Dialogues*”, The Association for the Advancement of Artificial Intelligence (AAAI Bridge on Logic & AI), 2026.
6. **Rajashik Datta**, Sanjan Baitalik, Sruti Das Choudhury, Arup Kumar Chattopadhyay, Amit Kumar Das, “Fuzzy Hypergraph Feature Association Map for High-Dimensional Feature Selection in Agriculture and Remote Sensing”, *International Journal of Fuzzy Systems*, 2026.
7. Sruti Das Choudhury, **Rajashik Datta**, Sanjan Baitalik, “Enhancing interpretability through clustering, explainable AI, and narrative visualization: applications in precision agriculture and healthcare patient segmentation”, *Information*, 2025.
8. Sanjan Baitalik, **Rajashik Datta**, Sanket Ghosh, Darothi Sarkar, Ayan Chaudhuri, “Machine Learning-Driven Insights For Stock Market Analysis And Trading”, *International Conference on Interdisciplinary Research in Technology and Management (IRTM 2024)*.
9. Sanket Ghosh, Sanjan Baitalik, **Rajashik Datta**, Darothi Sarkar, “The COVID-19 Shock: An Analysis Of Impacts And Responses Of Indian Stock Market”, *International Conference on Interdisciplinary Research in Technology and Management (IRTM 2024)*.
10. **Rajashik Datta**, Sanjan Baitalik, Sanket Ghosh, Saugata Ghosh, Swarnendu Ghosh, “Is Indian Financial Market Ready for Pandemics?”, *International Conference on Advancing Science and Technologies in Health Science (IEM-HEALS 2024) Book of Abstracts*.

Submitted

1. Sanjan Baitalik, **Rajashik Datta**, Arup Kumar Chattopadhyay, Amit Kumar Das, Amlan Chakraborty, “From Graphs to Hypergraphs: Submodular Coverage-Based Feature Selection on Intuitionistic Fuzzy Hypergraphs (SIFHFAM)”, *Pattern Recognition*, 2026.
2. Sanjan Baitalik, **Rajashik Datta**, Darothi Sarkar, Ayan Chaudhuri, “MiQ-MCP: Valid and Conditionally Robust Uncertainty Quantification for High-Frequency Financial Time Series via Mondrian Conformalized Quantile Regression” (GitHub: [MiQ-MCP](#)), *Computational Economics*, 2025.

Skills & Activities

Programming: Python, C, C++, Java, MATLAB, $\text{\LaTeX}$	ML/AI: PyTorch, TensorFlow, Scikit-learn, Transformers
XAI: SHAP, LIME	Data: Pandas, NumPy, SciPy
Databases: MySQL, PostgreSQL, MongoDB	Cloud: Google Cloud (Cloud Run/Compute), AWS (S3/EC2)
Visualization: Matplotlib, Seaborn, Plotly, Tableau	Tools: TensorBoard, MATLAB App Designer
Activities: GenSpark 1.0 Ideathon (Organizer; coordinated 50+ teams; shortlisted funded ideas), Jun–Aug 2025; IEM-ICDC 2025 (Conference volunteer; coordination & support), Apr 2025; Department of CSE, IEM (Assisted NBA accreditation documentation), Mar 2024	

Projects

Quantization-Aware Momentum | [GitHub](#)

1-bit Momentum optimizer with error feedback; matches full-precision Momentum SGD. Logistic regression ($n=4000$, $d=2000$, 5000 steps): train loss 2.809×10^{-3} vs signSGD 3.7614×10^{-2} ($13.39\times$ higher); remains 6–13 $\times$ better across weight decay sweeps.

Online Learning (VS-AdaGrad) | [GitHub](#)

Online learning for non-stationary time series via drift-aware, volatility-scaled AdaGrad; on piecewise AR(5) with 5 regimes ($T \approx 5000$, 10 seeds), improves regret proxy over AdaGrad by 18.4% (small drift) and 19.8% (medium), and beats tuned OGD by 23.7–63.8% across drift regimes.